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MID SEM REPORT

Numerical Investigation of Ammonia-Diesel Spray and Mixing Behaviour in Compression Ignition Engines

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1. Introduction

Compression-ignition (CI) engines are widely used in transportation, power generation, and heavy-duty applications because of their high efficiency and reliability. However, conventional diesel combustion is associated with emissions and dependence on petroleum-derived fuels. Ammonia has attracted attention as an alternative carbon-free fuel because it can be produced using renewable energy and does not contain carbon. In ammonia–diesel dual-fuel CI engines, ammonia can provide a major portion of the fuel energy while a smaller quantity of diesel acts as a pilot fuel for ignition.

Spray development and mixture formation are important stages in the operation of a CI engine. After injection, the liquid fuel undergoes atomization, breakup, evaporation, entrainment of surrounding gas, and mixing. The resulting spatial distribution of fuel and air strongly influences subsequent ignition and combustion. Parameters such as spray penetration, droplet size, evaporation rate, injection conditions, and ambient conditions therefore play an important role in the overall combustion process [1,3,4].

Computational fluid dynamics (CFD) provides a useful approach for investigating these processes because individual spray phenomena can be studied numerically under controlled conditions. Diesel spray modelling has been extensively investigated using Eulerian–Lagrangian approaches and different breakup, evaporation, and turbulence models [5–10]. However, ammonia and diesel possess significantly different physical and evaporation characteristics, making direct application of conventional diesel-spray understanding to ammonia–diesel systems insufficient. Recent studies have investigated ammonia sprays, ammonia–diesel spray interaction, and dual-fuel engine behaviour [2,11–15].

The present work therefore focuses on the numerical investigation of **ammonia–diesel spray and mixing behaviour in CI engines**. The initial stage of the work involves establishing and verifying a Lagrangian spray/parcel modelling framework in OpenFOAM. The framework will subsequently be extended towards diesel and ammonia–diesel spray configurations and used to investigate mixture formation under relevant operating conditions.

2. Literature Review

2.1 Diesel Spray Fundamentals

Fuel spray development is one of the most important processes governing mixture formation in CI engines. Following injection, the liquid jet undergoes atomization and breakup while entraining the surrounding gas. The resulting spray structure is commonly characterized using parameters such as spray penetration, spray cone angle, droplet size, and spatial fuel distribution.

Spray penetration is influenced by injection conditions, injector geometry, ambient gas density, and fuel properties. Fundamental studies have established relationships for describing the penetration of diesel sprays, while subsequent work has interpreted spray development using characteristic breakup length and time scales [3,4]. The development of the spray is not governed by a single process; primary breakup, secondary breakup, droplet motion, evaporation, entrainment, and mixing occur simultaneously [1].

These processes determine how rapidly the injected fuel occupies the combustion chamber and mixes with the surrounding air. Consequently, understanding conventional diesel spray behaviour provides an important basis for developing numerical models for alternative-fuel CI engines.

2.2 Numerical Modelling of Diesel Spray

Numerical spray simulations commonly employ an **Eulerian–Lagrangian framework**, in which the surrounding gas is solved using an Eulerian computational mesh while the injected liquid is represented by discrete Lagrangian parcels. The parcels represent groups of droplets having specified physical properties and initial conditions.

Breakup modelling is required to represent the transformation of the injected liquid jet into smaller droplets. Models based on Kelvin–Helmholtz (KH), Rayleigh–Taylor (RT), and combined KH–RT mechanisms are widely used for diesel spray simulations [5–8]. Following breakup, droplet motion is influenced by drag and other forces, while evaporation depends on droplet temperature, fuel properties, and surrounding gas conditions.

Turbulence modelling also affects predicted spray development and fuel–air mixing. Both RANS and LES approaches have been investigated for diesel spray simulations, with the predicted spray structure depending on the treatment of turbulent flow and spray–turbulence interaction [10]. Reviews of liquid atomization and spray modelling have highlighted the strongly coupled nature of injection, breakup, droplet dynamics, evaporation, and mixing [6,9].

These numerical approaches provide the foundation for the present work. However, their application to ammonia–diesel systems requires consideration of the different physical and evaporation characteristics of ammonia and diesel.

2.3 Ammonia Spray Characteristics

Ammonia differs significantly from diesel in its thermophysical properties and evaporation behaviour. Studies of high-pressure liquid ammonia injection have shown that ammonia spray development and evaporation can differ from conventional diesel sprays under comparable conditions [13].

The development of an ammonia spray is affected by parameters such as ambient pressure and temperature, injection conditions, and fuel properties. Because ammonia can undergo significant phase change during spray development, evaporation and the resulting vapour distribution are particularly important when analysing mixture formation.

Recent investigations have therefore examined high-pressure ammonia spray characteristics under conditions relevant to ammonia–diesel CI engines [13]. These studies indicate that conventional diesel spray correlations and models cannot necessarily be transferred directly to ammonia without considering the differences in fuel properties and phase-change behaviour.

2.4 Ammonia–Diesel Spray Interaction

In an ammonia–diesel dual-fuel CI engine, the two fuels may undergo different spray and evaporation processes before combustion. Their interaction depends on the injection configuration, relative spray orientation, injection timing, fuel properties, and surrounding gas conditions.

Numerical and experimental investigations have shown that spray interaction can influence the spatial distribution of the two fuels and consequently affect mixture formation and subsequent combustion [11]. The behaviour of high-pressure ammonia sprays in ammonia–diesel engines has also been investigated, demonstrating the importance of understanding ammonia spray development together with the diesel pilot [13].

The resulting mixture is therefore determined not only by the individual spray characteristics of ammonia and diesel but also by their interaction, entrainment, evaporation, and mixing with the surrounding air. Understanding these processes provides a link between injection conditions and the subsequent combustion behaviour of dual-fuel CI engines.

2.5 Numerical and Experimental Studies on Ammonia–Diesel Engines

Recent studies have investigated ammonia–diesel dual-fuel engines using both experimental and numerical approaches. Engine-level investigations have examined performance, combustion, and emissions when ammonia is combined with diesel as the pilot fuel [2,12,15].

Numerical studies have also investigated ammonia–diesel combustion and spray behaviour under engine-relevant conditions. Such studies demonstrate the importance of accurately representing fuel injection and mixture formation before attempting to predict combustion and emissions. Investigations involving direct injection and combustion have further shown the coupled nature of spray, mixing, combustion, and ammonia-slip behaviour [12,15].

Although these studies provide important information about ammonia–diesel engine operation, complete engine simulations involve several simultaneously coupled physical and chemical processes. A focused investigation of spray and mixing behaviour can therefore provide a more controlled way of understanding the early stages of mixture formation.

3. Research Gap

Diesel spray development and numerical spray modelling have been extensively investigated, with established approaches for representing spray penetration, breakup, droplet motion, evaporation, turbulence, and fuel–air mixing [1,3–10]. These studies provide a strong foundation for numerical investigation of liquid-fuel sprays.

In comparison, ammonia-based dual-fuel CI systems are less extensively studied from the perspective of **spray and mixture formation alone**. Recent investigations have examined liquid ammonia sprays, ammonia–diesel spray interaction, and complete ammonia–diesel engine operation [2,11–15]. These studies indicate that the different physical and evaporation characteristics of ammonia and diesel can influence fuel distribution and subsequent combustion.

However, the connection between individual spray processes and the resulting spatial and temporal distribution of ammonia and diesel during mixture formation requires further investigation. Complete engine simulations also involve combustion and emissions processes, making it difficult to isolate the contribution of spray and mixing phenomena.

Therefore, the present work focuses on developing a numerical framework for investigating **ammonia–diesel spray development and mixing behaviour**. The initial work concentrates on establishing the numerical framework and verifying its basic operation, followed by investigation of diesel and ammonia–diesel spray configurations under controlled conditions.

4. Objectives

The main objective of the present work is to **numerically investigate the spray and mixing behaviour of ammonia–diesel fuel systems under CI-engine-relevant conditions**.

The specific objectives are:

1. To study the fundamental development of liquid-fuel sprays, including spray penetration and spatial evolution.
2. To develop a numerical framework for injection, breakup, droplet motion, evaporation, and spray interaction.
3. To investigate the different spray and mixing characteristics of ammonia and diesel.
4. To analyse the spatial and temporal distribution of ammonia and diesel during mixture formation.
5. To investigate the influence of selected injection and operating parameters on spray and mixing behaviour.
6. To establish a numerical foundation for subsequent investigation of combustion and emissions in ammonia–diesel CI engines.

5. Methodology

The numerical investigation is being developed using an OpenFOAM-based CFD framework. The initial stage focuses on the liquid-phase parcel and spray processes before introducing the additional complexity of combustion and emissions.

The planned methodology consists of the following stages:

Computational domain → Mesh generation → Initial/boundary conditions → Fuel injection → Spray breakup → Droplet motion and evaporation → Fuel–air mixing → Post-processing → Validation

5.1 Computational Domain and Mesh

A controlled computational domain will initially be used to investigate spray development without the additional complexity of a complete engine geometry. The mesh will be selected to provide adequate resolution of the spray region while maintaining reasonable computational cost.

5.2 Injection and Operating Conditions

The fuel injection conditions will be defined using parameters such as injection pressure, injection duration, injection velocity, injector characteristics, fuel properties, and ambient

pressure and temperature. These parameters will subsequently be varied during the parametric investigation.

5.3 Spray Modelling

The injected liquid will be represented using Lagrangian parcels. Appropriate models will be incorporated to represent primary and secondary breakup, droplet motion, and interaction with the surrounding gas. The choice of models will depend on the fuel and spray configuration being investigated.

5.4 Evaporation and Mixing

Heat and mass transfer between the droplets and surrounding gas will be considered to represent evaporation. The resulting vapour distribution will be analysed together with the liquid-phase spray to determine the development of the fuel–air mixture.

5.5 Post-processing

The numerical results will be analysed using parameters such as:

- spray penetration,
- droplet/parcel velocity,
- liquid and vapour fuel distribution,
- evaporation behaviour,
- spatial distribution of ammonia and diesel, and
- fuel–air mixing characteristics.

Where suitable experimental or published numerical results are available, the predicted spray characteristics will be compared with reported data for validation.

6. Preliminary Numerical Investigation

6.1 Computational Framework

As an initial verification step, a water-droplet Lagrangian parcel simulation was performed using OpenFOAM-14. This case was not intended to represent ammonia–diesel spray behaviour; rather, it was used to verify the basic numerical framework required for subsequent fuel-spray simulations, including Lagrangian parcel tracking, droplet forces, heat transfer, and evaporation.

The `multicomponentFluid` solver was used, with a single liquid water parcel introduced into a gaseous computational domain. The simulation therefore represents a preliminary verification case rather than a direct ammonia–diesel engine simulation.

6.2 Computational Domain and Numerical Setup

A three-dimensional computational domain of dimensions **0.1 m × 0.1 m × 0.01 m** was used and discretized into **125 computational cells**.

A single liquid parcel was introduced at the position

(0.05, 0.05, 0.005) m

with an initial velocity of **0 m/s**. The initial parcel diameter was **100 µm**, with an initial mass of **5.23599×10^{-10} kg**.

For this verification case, the liquid phase was represented by **water**. The parcel was subjected to gravitational and drag forces. Heat transfer between the parcel and surrounding gas was modelled using the **Ranz–Marshall model**, and liquid evaporation was activated.

The simulation was performed for **0.5 s** using an initial time step of **1×10^{-3} s**.

6.3 Preliminary Numerical Results

The simulation completed successfully and demonstrated Lagrangian parcel tracking together with heat transfer and phase-change calculations.

The initial liquid mass of the parcel was **5.23599×10^{-10} kg**, while the final liquid mass after 0.5 s was **3.29855×10^{-11} kg**. The reported phase-change mass was approximately **4.90614×10^{-10} kg**. The final parcel temperature was approximately **319.73 K**.

The reduction in liquid-phase mass indicates substantial phase change during the simulation. The numerical results therefore confirm that the implemented framework can track a liquid parcel while accounting for droplet forces, heat transfer, and evaporation.

Table 1. Summary of preliminary numerical simulation

Parameter	Value
Computational Domain	0.1 × 0.1 × 0.01 m
Number of Cells	125
Initial parcel diameter	100 μm
Initial parcel mass	5.23599×10^{-10} kg
Initial parcel velocity	0 m/s
Simulation time	0.5 s
Final liquid mass	3.29855×10^{-11} kg
Phase-change mass	4.90614×10^{-10} kg
Final parcel temperature	≈319.73 K

6.4 Relevance to Present Work

The preliminary simulation represents a **verification of the numerical framework rather than a direct ammonia–diesel engine result**. The purpose of the case was to establish the procedure for setting up a Lagrangian liquid-phase calculation and to verify parcel tracking, droplet forces, heat transfer, and evaporation.

The next stage will extend this framework towards transient fuel injection and spray development under CI-engine-relevant conditions. Diesel will initially be considered as a reference fuel, after which the framework will be extended to ammonia–diesel spray interaction and mixing.

7. Proposed Future Work

Based on the preliminary numerical investigation, the following work is proposed:

1. **Development of a transient fuel-spray model:**
   Extend the present parcel-based framework to represent fuel injection through an injector under CI-engine-relevant conditions.
2. **Diesel spray investigation:**
   Establish a diesel reference case and investigate spray penetration, structure, droplet behaviour, evaporation, and fuel distribution.
3. **Spray breakup and evaporation modelling:**
   Implement and assess appropriate breakup and evaporation models for the selected fuel and operating conditions.
4. **Ammonia–diesel spray investigation:**
   Extend the framework to ammonia–diesel injection and investigate the interaction and mixing of the two fuels.
5. **Parametric study and validation:**
   Study the influence of selected parameters such as injection pressure, injection timing, ambient conditions, and fuel-injection conditions, and compare suitable results with published experimental or numerical data.
6. **Extension towards combustion and emissions:**
   After establishing a reliable spray and mixing framework, extend the numerical model towards combustion and emission analysis of ammonia–diesel dual-fuel CI engines.

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